

Week One

Lecture: "Statistical Concepts: Part 1"

There is a mistake in the standard deviation formula that was propagated throughout the lecture. The square root symbol should include the denominator of the expression (in the version in the slides, the square root is only applied to the numerator). Below are fixed versions of the equation for the population, sample, and standard error.

Standard Deviation of Populations

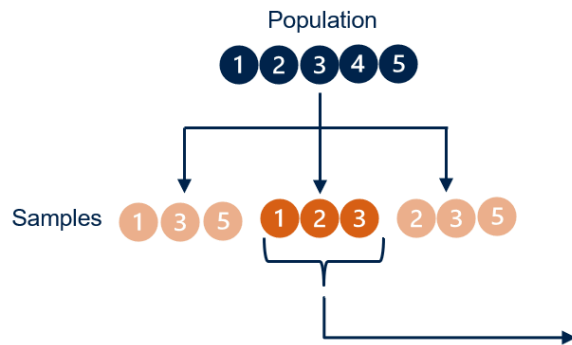
Population
1 2 3 4 5



Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

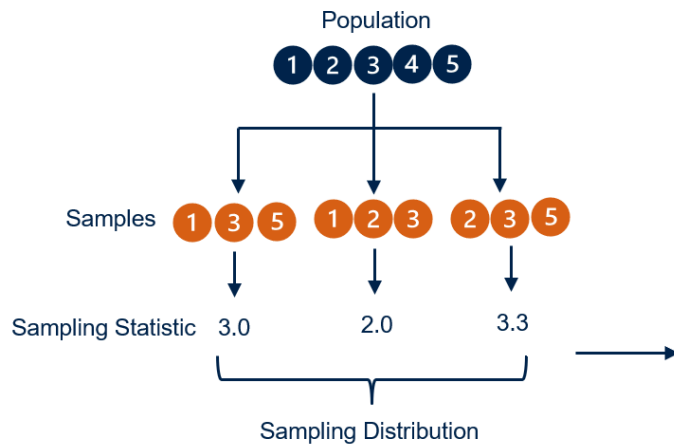
Standard Deviation of Samples



Sample Standard Deviation

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Standard Error of Samples



Standard Error

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$